Diversity and Inclusion in Clinical Trials

Go to diversity and inclusion in clinical trials scientific resources

Our health is a combination of physical and mental well-being, which is affected by our behavior, biology, environment, societal policies, and importantly, our lived experiences. The lived experiences of people in the United States vary based on their race and ethnicity, socioeconomic status (SES), geographic location, sexual orientation, gender identity, and other sociodemographic characteristics.

Lived experiences also need to be understood in the context of the individual and structural social determinants of health.

How and where we live, learn, work and play, and our access to high quality health care, healthy foods, and quality education can enhance our health outcomes.

Similarly, negative experiences and exposures, such as pollution, violence, and structural racism and discrimination, can negatively affect our health.

Our health status reflects the interwoven effects of such factors.

A clinical trial (https://grants.nih.gov/policy/clinical-trials/definition.htm) is a type of clinical research (https://www.nih.gov/health-information/nih-clinical-research-trials-you/basics) that evaluates the effects of intervention(s), including drugs, devices, surgeries, diets, behavioral approaches, and lifestyle interventions, on health-related biomedical or behavioral outcomes.

To account for the diverse lived experiences and exposures of various populations, clinical research should be appropriately **inclusive** of racial and ethnic minority groups, as well as other populations experiencing health disparities, including sexual and gender minority or socioeconomically disadvantaged populations.

Why Are Clinical Trials Important?

Clinical trials can:

- Determine if a new intervention is safe, works better, and/or has fewer side effects than an existing treatment or intervention.
- Examine ways to detect a disease early, when it is potentially more treatable, or ways to prevent a health problem altogether.
- Evaluate ways to improve the quality of life of people who have an illness or chronic medical condition.
- Include testing of behavioral, social, environmental, and structural interventions.

Participating in clinical trials is voluntary. People volunteer to participate in clinical trials for a variety of reasons.

- One of the most common reasons is altruism—the opportunity to contribute to science and the common good and/or help those with similar health issues.
- People may volunteer when it allows them to receive an experimental intervention for life-threatening or disabling disease where no standard treatments are available or were already tried without success.
- New interventions (e.g., weight loss or tobacco cessation interventions) that haven’t yet been approved by the U.S. Food and Drug Administration (FDA) may be tested for common conditions to understand if the intervention might help a condition in situations where current treatments or interventions don’t exist, don’t work well, or have unwanted side effects, or provide symptomatic relief, but offer no cure.

Medical Equality by Arielle Clark. Clark's illustration was a winning teen entry in NIMHD's Envisioning Health Equity Art Challenge 2020 (/programs/edu-training/art-challenge/index.html).

The Importance of Diversity & Inclusion in Clinical Trials

People may experience the same disease differently. It's essential that clinical trials include people with a variety of lived experiences and living conditions, as well as characteristics like race and ethnicity, age, sex, and sexual orientation, so that all communities can benefit from scientific advances.

Factors that can influence the risk and likelihood of developing a disease, experiencing a long-term health outcome, and responding to treatment include (but are not limited to):

- Age
- Biological sex
- Pregnancy status
- Life experiences (negatives, such as psychosocial stress and lack of basic resources, or positives, such as educational and employment opportunities)
- Unhealthy behaviors (e.g., substance use, sedentary lifestyle, overeating, risky sexual activity)
- Health-promoting behaviors (e.g., adequate sleep, obtaining recommended preventive services, physical activity, healthy eating)
- Environmental conditions (e.g., pollution, access to health care or healthy foods, neighborhood segregation)
- Genetic variation and geographic ancestry
- Underlying medical problems or presence of comorbidities (i.e., additional diseases or conditions)

Historically, clinical trials did not always recruit participants who represented the individuals most affected by a particular disease, condition, or behavior. Often, these clinical trials relied almost exclusively on White male study participants. This shortcoming has created gaps in our understanding of diseases and conditions, preventive factors, and treatment effectiveness across populations. These gaps in knowledge can impede the quality of health care decision making, ability to counsel people on ways to reduce their risk, optimal treatment responses, and even the development of more effective medications or interventions.

Clinicians and researchers should carefully consider the inclusion or exclusion criteria for their clinical trials. For example, a clinical trial excluding participants with high blood pressure or other comorbidities may end up excluding many people over 65 years old, who are more likely to have these conditions. The trial may then underrepresent certain groups in the study and make the results less applicable to groups who may benefit the most from the findings.

Real-World Examples of the Need for Inclusion in Clinical Trials

Understanding COVID-19 Disparities

During the early stages of the pandemic, Coronavirus disease 2019 (COVID-19) disproportionately affected (https://covid.cdc.gov/covid-data-tracker/#demographicsovertime) racial and ethnic minority populations, including African American, Hispanic/Latino, American Indian/Alaska Native, and Native Hawaiian and Pacific Islander (https://www.kff.org/report-section/covid-19-racial-disparities-in-testing-infection-hospitalization-and-death-analysis-of-epic-patient-data-issue-brief/) (/disclaimer/) population groups, with increased cases, hospitalizations and deaths.

It was critical that COVID-19 vaccine trials (https://covid19.nih.gov/treatments-and-vaccines/clinical-trials) included sufficient representation across population groups to better understand vaccine effectiveness in populations who vary on environmental exposures and other lived experiences. By using inclusive recruitment practices (https://covid19community.nih.gov/resources/ensuring-inclusion) in COVID-19 clinical trials, researchers have been able to show that vaccine safety and efficacy are similar across all racial and ethnic populations. Engaging diverse populations

(https://covid19community.nih.gov/sites/default/files/Promoting_Trust_and_Inclusion_in_COVID-19_Research.pdf) in planning and implementing such trials can also help increase public confidence (https://www.kff.org/racial-equity-and-health-policy/issue-brief/racial-diversity-within-covid-19-vaccine-clinical-trials-key-questions-and-answers/) (/disclaimer/) that the vaccine is safe and effective.

Understanding Asthma Disparities

Asthma disparities are intricately linked with the environment. Living in a city (/news-events/features/community-health/location-matters.html) may increase exposure to air pollution and risk for developing asthma. Exposure to tobacco smoke, chronic social stress (/news-events/features/biological-behavioral-siences/asthma.html), or unhealthy diets (/news-events/research-spotlights/diet-asthma.html) may also influence asthma risk or severity. Thus, it is vital for clinicians and researchers to consider where patients live, what they eat, and how they feel—as well as characteristics like race, ethnicity, socioeconomic status, and age—to get a more thorough understanding of their patients' experience with asthma symptoms and identify the best preventative strategies or treatment options.

Inclusive Participation in Clinical Trials Benefits Scientific Discovery

NIH is committed to inclusivity in clinical trial research (https://orwh.od.nih.gov/womens-health/clinical-research-and-trials). It is essential to have a wide range of people from different communities participate in clinical trials to reduce biases, promote social justice and health equity, and produce more innovative science. Below is a list of topics and examples to illustrate the important role of inclusive participation in clinical trial research.

Countering Mistrust in Clinical Research

Historical atrocities and incidents have engendered mistrust in clinical research and medical institutions.

Investigators conducting the U.S. Public Health Service Syphilis Study at Tuskegee (https://www.cdc.gov/tuskegee/index.html) between 1932 and 1972 did not explain the study's risks and obtain formal agreements (called informed consent (https://www.cancer.gov/about-cancer/treatment/clinical-trials/patient-safety/informed-consent)) from the African American men who were its participants. The researchers wanted to study the effects of untreated syphilis and withheld penicillin treatment when it became available in 1945, which would have helped the 399 study participants with the disease. Only when news leaked of the study in 1972 did their unethical and discriminatory behavior come to light. Their actions caused preventable illness and death in study participants and their families.

In 2003, members of the Havasupai Tribe in Northern Arizona learned (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5310710/) that DNA samples given in the early 1990's for a diabetes research study were later being used for additional research on ethnic migration, schizophrenia, and other unrelated genetic studies. The informed consent form from the original study did not ask participants for their permission to use these samples for these other analyses. The researchers failed to obtain their consent for use of their data and specimens for other research purposes.

The failings of the Syphilis Study at Tuskegee contributed to the creation of the Belmont Report (https://www.hhs.gov/ohrp/regulations-and-policy/belmont-report/index.html) in 1976, which addresses ethical issues in research with human participants. It outlines basic ethical principles and essential guidelines to protect human research participants and ensure safety in clinical trial research.

Today, Institutional Review Boards are responsible for reviewing all studies involving humans for compliance with these guidelines and reports of any study protocol violations. In recent years, people from racial and ethnic minority communities and other populations experiencing health disparities have become more willing to participate in clinical research (https://journals.plos.org/plosmedicine/article?id=10.1371/journal.pmed.0030019) (/disclaimer/). Developing trust with communities who have been marginalized is best achieved through meaningful partnerships between researchers and community members in planning and carrying out studies with their input.

Inclusion of Women and People from Racial and Ethnic Minority Groups in Clinical Trials

The NIH Revitalization of Act of 1993 (https://www.govinfo.gov/content/pkg/USCODE-2011-title42/pdf/USCODE-2011-title42-chap6A-subchapIII-partH-sec289a-2.pdf) was signed into law, authorizing NIH to continue its mission and importantly establishing guidelines for the inclusion of women and persons from racial and ethnic minority populations (https://grants.nih.gov/policy/inclusion/women-and-minorities.htm) in clinical research. The goal of this law, and other guidelines, is for clinical trial participants to adequately reflect the diversity of the real-world population, so that researchers can determine whether the variables being studied affect women or members of any racial and ethnic population group. This helps ensure that research findings are generalizable to the entire population.

NIH efforts toward research inclusion remain at the forefront of clinical research policy (https://orwh.od.nih.gov/womens-health-research/clinical-research-trials/nih-inclusion-policies). Recent activities include the publicly available NIH Research, Conditions and Disease Categorization Inclusion Statistics Report (https://report.nih.gov/RISR/#/), which provides data on human research participation in NIH clinical research studies by race, ethnicity, and sex/gender. Additionally, in 2017, NIH updated its policy on the inclusion of women and people from racial and ethnic minority populations (https://grants.nih.gov/grants/guide/notice-files/NOT-OD-18-014.html) with a requirement that "recipients conducting applicable NIH-defined Phase III clinical trials (https://grants.nih.gov/grants/glossary.htm#NIH-DefinedPhaseIIIClinicalTrial) ensure results of valid analyses by sex/gender, race, and/or ethnicity are submitted to Clinicaltrials.gov (https://clinicaltrials.gov/)." See NIH Inclusion Outreach Toolkit: How to Engage, Recruit, and Retain Women in Clinical Research (https://orwh.od.nih.gov/toolkit) for more information.

Inclusion of Sexual and Gender Minority Populations

Until recently, health care systems and epidemiological surveys often didn't ask sexual orientation and gender identity questions to consider inclusion of sexual and gender minority (SGM) persons. This has made it difficult to know if individuals within SGM populations are represented in clinical research studies in significant numbers to make results representative for them. This lack of knowledge can influence patient-clinician communication (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6657380/) and can result in fewer health screening or treatment opportunities (https://pubmed.ncbi.nlm.nih.gov/26789508/).

Inclusion by Socioeconomic Status (SES)

An individual's SES is a major predictor of health outcomes, because it can impact access to health care, nutritious foods, prescription medications, and other resources for healthy living. Yet, SES measures (i.e., education and income level) are not collected routinely and reported in clinical trials.

In an analysis of all randomized clinical trials published in 2015 and 2019 in the *Journal of the American Medical Association, The Lancet*, and the *New England Journal of Medicine*, study investigators reported that less than 15% of studies reported on the SES (https://gcc02.safelinks.protection.outlook.com/?url=https%3A%2F%2Fjamanetwork.com%2Fjournals%2Fjamanetworkopen%2Ffullarticle%2F2780381&data=05%7C01%7Cmonica.hooper%40nih.gov%7Cc0efbb43a4aa4 (/disclaimer/) of trial participants. Lack of data collection and reporting on SES measures make it difficult to generalize research findings to all SES groups or to tailor interventions (e.g., new medications or other treatment interventions) to people with lower SES who may not be able to access or maximize the benefits of clinical trial outcomes. In addition, limited access to socioeconomic resources may pose a barrier to participation in clinical trials.

To ensure the inclusion and representation of participants across different SES levels in clinical trials, researchers should use appropriate data collection and reporting protocols. For example, NIMHD supported a social determinants of health collection in the PhenX Toolkit (https://www.phenxtoolkit.org/collections/view/6) (/disclaimer/) that includes established instruments for conducting research with human participants, such as clinical trials.

Researchers should also design their studies and provide resources to make it easier for people with lower SES to participate in clinical trials, such as offering convenient locations and hours of operation, childcare services, and transportation vouchers.

Data Collection and Reporting: Unmasking Hidden Truths

When scientists combine information from individual research participants, this is called data aggregation. Data aggregation is an important part of the research process that protects the anonymity of research volunteers and strengthens the statistical analysis of the study. However, aggregation of demographic data, including race and ethnicity, can also mask important differences in health risks or outcomes for specific subpopulations.

For example, many prior studies on the health of Asian Americans have not always examined differences by nationality. A recent study found that among Filipino, Vietnamese, Chinese, Japanese, and Korean American adults living in California, categorizing all participants as "Asian American" masked at least one health disparity (https://bmcpublichealth.biomedcentral.com/articles/10.1186/s12889-019-7683-3) (/disclaimer/) for each subpopulation.

Clinicians and researchers must take care to define as best as possible the clinical trial sample in their studies and consider whether their findings can be generalized across population groups, including consideration for differences in lived experiences.

NIMHD is studying and addressing issues related to diversity and inclusion in clinical trials through a variety of strategic goals, funding initiatives, and educational materials:

NIMHD Content & Resources

NIH Minority Health & Health Disparities Strategic Plan 2021-2025

The National Institutes of Health Minority Health and Health Disparities Strategic Plan 2021-2025 (/about/strategic-plan/) was developed by NIMHD, in collaboration with all the NIH Institutes, to outline NIH's commitment to improving minority health and reducing health disparities. Two of the Strategic Plan's major goals focus on issues of diversity and inclusion in NIH-funded research and in clinical trials.

- Goal 7: Ensure appropriate representation of minority and other population experiencing health disparities in NIH-funded research (/about/strategic-plan/nih-strategic-plan-research-sustaining-activities.html#section7)
- Goal 8: Promote evidence-based community engagement, dissemination, and implementation of minority health and health disparities research best practices (/about/strategic-plan/nih-strategic-plan-outreach-collaboration-dissemination.html#section8)

The Strategic Plan's Leap Forward Research Challenges (/about/strategic-plan/nih-strategic-plan-leap-forward-research-challenge.html) are goals that aim to redefine the science of minority health and health disparities. These goals include:

- Increase the overall proportion of participants from diverse populations included in NIH-funded clinical research to 40 percent by 2030 and within specific major disease categories.
- Increase the diversity of institutions conducting genomic research and training by investing in faculty at such institutions, along with curriculum-building partnerships, to accelerate workforce development in underrepresented and under-resourced communities within the next 10 years.

NIMHD Articles

Research Features

- Q&A: Tuskegee University Researchers Embrace History and Community (/news-events/features/community-health/tuskegee-university.html)
- Patient Navigators: The Missing Link to Increasing Minority Participation in Cancer Clinical Trials (/news-events/features/clinical-health-services/patient-navigators.html)
- The Center for Asian Health Engages Communities in Research to Reduce Asian American Health Disparities (/news-events/features/training-workforce-dev/center-asian-health.html)

NIMHD Insights Blog

- The Sweetness of Our Ancestors: Thoughts on Diabetes, Genetics, and Ethnic Diversity in Research (https://blog.nimhd.nih.gov/archives-2021/news_feed/the-sweetness-of-our-ancestors-thoughts-on-diabetes-genetics-and-ethnic-diversity-in-research) by Larissa Avilés-Santa, M.D., M.P.H.
- The Power of Trust and Truth (https://blog.nimhd.nih.gov/archives-2020/news_feed/the-power-of-trust-and-truth) by NHLBI Director Gary H. Gibbons, M.D. and NIMHD Director Eliseo J. Pérez-Stable, M.D.
- Communicating the Value of Race and Ethnicity in Research (https://blog.nimhd.nih.gov/archives-2018/news_feed/communicating-the-value-of-race-and-ethnicity-in-research) by NIMHD Director Eliseo J. Pérez-Stable, M.D.
- Reducing Health Disparities and Enhancing Diversity in Aging Research (https://blog.nimhd.nih.gov/archives-2018/news_feed/guest-blog-post-reducing-health-disparities-and-enhancing-diversity-in-aging-research) by National Institute on Aging Director Richard J. Hodes, M.D.

NIMHD Funding Resources & Opportunities

Funding Opportunity Announcements

Clinical trials and other forms of clinical research focused on populations experiencing health disparities are vital to improving our understanding of the causes of many diseases that affect these populations. New insights from such studies can transform medical practice and health promotion in targeted populations. See Active NIMHD Funding Opportunities (/funding/nimhd-funding/active_foa.html) for more information.

NIMHD-Supported Research Projects

NIMHD sponsored a Funding Opportunity Announcement (RFA-MD-11-005 (https://grants.nih.gov/grants/guide/rfa-files/rfa-md-11-005.html)) to support projects that identified evidence-based strategies to increase enrollment of racial and ethnic minority persons into clinical trials. Below are examples of some of these projects with links to published findings or guidelines for best practices. Projects included:

- **National Initiative for Minority Involvement in Neurological Clinical Trials (NIMICT) (U24MD006961 (https://reporter.nih.gov/search/HwonVmgsHEmo632y3GQRCA/projects?shared=true))**
  The goal of this project was to identify and evaluate strategies to promote recruitment and retention of racial and ethnic minority persons in neurological clinical trials. The NIMICT project (https://nimict.com/) (/disclaimer/) generated toolkits, manuals, educational videos, and online training modules to guide investigators through the process of designing inclusive clinical trials. Project investigators:
  - Worked with StrokeNet (https://www.nihstrokenet.org/) (/disclaimer/), NIH's network of clinical trials addressing stroke, to ensure that all funded trials have acceptable racial and ethnic minority participant recruitment plans.
  - Identified barriers and strategies (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5547022/) for recruitment of racial and ethnic minority persons in neurological research.
  - Examined barriers and practices (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5526461/) for recruitment and retention of participants in stroke clinical trials.
- **Enhancing Minority Participation in Clinical Trials (EMPaCT): Phase II (U24MD006970 (https://reporter.nih.gov/search/oBXtyidaq0GrrZhTE-u4IQ/projects?shared=true))**
  The goal of this project was to test strategies to increase racial and ethnic minority patient enrollment in cancer clinical trials. The work for this project was conducted across five National Cancer Institute Comprehensive Cancer Centers. The investigators:
  - Generated strategies that facilitate patient engagement in cancer care, including understanding institutional barriers to recruitment and utilizing patient navigators (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4957258/) to facilitate the enrollment of racial and ethnic minority individuals into cancer clinical trials.
  - Examined bias and stereotyping (https://pubmed.ncbi.nlm.nih.gov/32147815/) and the impact on recruitment to cancer clinical trials.
  - Identified needed training to increase cultural awareness of clinical and research professionals to help recruitment and retention in cancer trials
  - Assessed the motivations of clinical and research staff (https://pubmed.ncbi.nlm.nih.gov/27826818/) for including racial and ethnic minority populations in cancer clinical trials.

NIMHD-Supported, NIH-Wide Initiatives

- The NIH RePORT on Inclusion of Women and Minorities in Clinical Research (https://report.nih.gov/research/inclusion-women-and-minorities-clinical-research) is a database that contains inclusion information on NIH-funded clinical research since 1994.
- NIH Clinical Research Trials and You (https://www.nih.gov/health-information/nih-clinical-research-trials-you) features educational resources and information about diversity in research, including policies and guidance on the inclusion of women and racial and ethnic minority populations

(https://grants.nih.gov/policy/inclusion/women-and-minorities.htm) and population groups across the lifespan (https://grants.nih.gov/policy/inclusion/lifespan.htm).

- Diversity is a core value (https://allofus.nih.gov/about/diversity-and-inclusion) of the NIH *All of Us* Research Program (https://allofus.nih.gov/), which stresses the importance of the participation of people of different races, ethnicities, age groups, regions across the country, gender identities, sexual orientations, socioeconomic backgrounds, disabilities, and health status.
- NIH Inclusion Data by Research and Disease Category Now Available – NIH Extramural Nexus (https://nexus.od.nih.gov/all/2019/05/06/nih-inclusion-data-by-research-and-disease-category-now-available/)
- The National Institute on Aging's Recruiting Older Adults in Research (ROAR) Toolkit (https://www.nia.nih.gov/health/recruiting-older-adults-research-roar-toolkit).
- The National Institute of Mental Health's Points to Consider about Recruitment and Retention While Preparing a Clinical Research Study (https://www.nimh.nih.gov/funding/grant-writing-and-application-process/points-to-consider-about-recruitment-and-retention-while-preparing-a-clinical-research-study).
- The National Center for Advancing Translational Sciences Trial Innovation Network (https://ncats.nih.gov/ctsa/projects/network).
- The Community Engagement Alliance (CEAL) Against COVID-19 Disparities provides resources to promote and ensure diversity and inclusion in vaccine and diagnostic testing (https://covid19community.nih.gov/resources/ensuring-inclusion).

Additional Resources

- The Food and Drug Administration's Clinical Trial Diversity webpages (https://www.fda.gov/consumers/minority-health-and-health-equity/clinical-trial-diversity) include resources such as:
    - Clinical Trial Diversity Fact Sheet (https://www.fda.gov/media/106965/download)
    - Enhancing the Diversity of Clinical Trial Populations—Eligibility Criteria, Enrollment Practices, and Trial Designs Guidance for Industry (https://www.fda.gov/regulatory-information/search-fda-guidance-documents/enhancing-diversity-clinical-trial-populations-eligibility-criteria-enrollment-practices-and-trial)
- Improving Representation in Clinical Trials and Research: Building Research Equity for Women and Underrepresented Groups (https://nap.nationalacademies.org/catalog/26479/improving-representation-in-clinical-trials-and-research-building-research-equity) (/disclaimer/) | The National Academies Press

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Related

NIH Inclusion Policies (https://orwh.od.nih.gov/womens-health-research/clinical-research-trials/nih-inclusion-policies)

FDA Statement on Diversity in Clinical Trials (https://www.fda.gov/news-events/press-announcements/fda-offers-guidance-enhance-diversity-clinical-trials-encourage-inclusivity-medical-product)

NIH UNITE Initiative (https://www.nih.gov/ending-structural-racism/unite)

NIH CEAL: Diversity & Inclusion (https://covid19community.nih.gov/resources/ensuring-inclusion)

Understanding Health Disparities: Structural Racism and Discrimination (/resources/understanding-health-disparities/srd.html)

Learn More

NIMHD Fact Sheet (PDF, 90 KB) (/docs/about-nimhd-factsheet.pdf)

Read about what is
happening at NIMHD at the News and Events (/news-events/) section

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